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Research Division of Antarctic Astronomy and Radio Astronomy
Research of molecular gas in the Milky Way Group

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Experience

2020.02-2026.01, Assistant Researcher, PMO, CAS
2015-present, Scientific Group Member, Milky Way Imaging Scroll Painting Project, PMO
2010-2012, Physics Teacher, Changji No.1 High School, Xinjiang Autonomous Region

Education

2020, Ph.D. Astrophysics, University of Chinese Academy of Sciences
2010, B.S. Physics, Shannxi Normal University
Ph.D. Thesis title: An Observational Study of the Structure and Physical Properties of Molecular Clouds via Cases and a Sample

Research Interest

Turbulence in the Interstellar Medium
Structure and Dynamics of Molecular Clouds
Star Formation

Grants

As Principal Investigator (PI)

- **National Natural Science Foundation of China (NSFC), Youth Program (No. 12303033)**, *“Study on the Dominance and Fundamental Properties of Turbulence in Galactic Molecular Clouds”*
Jan. 2024 – Dec. 2026
- **PMO Internal Research Grant, Grants for Junior Talents (No. E2ZC201111)**, *“Study on the Column Density Structures of Galactic Molecular Clouds”*
Sep. 2022 – Aug. 2023

As Participant

- **National Key R&D Program of China (No. 2023YFA1608000)**, *“Study on the Distribution and Structure of Molecular Gas in the Milky Way”*
Sub-project: “Distribution and Properties of Molecular Clouds”
Dec. 2023 – Nov. 2028
- **National Natural Science Foundation of China (NSFC), General Program (No. 12473026)**, *“Star Formation from the Perspective of Complex Systems Science: A Statistical Study of Star Formation Rates in Galactic Molecular Clouds”*
, Jan. 2025 – Dec. 2028
- **NSFC General Program, (No. 11973091)**, *“Multiscale Analysis of Molecular Cloud Structures and Physical Properties”*
Jan. 2020 – Dec. 2023
- **NSFC General Program, (No. 11973090)**, *“Ring Structures and Formation Mechanisms of Protoplanetary Disks”*
Jan. 2020 – Dec. 2023
- **Jiangsu Provincial Natural Science Foundation, Youth Program (No. BK20181513)**, *“Numerical Modeling of Chemical Differentiation in Star-Forming Regions”*
Jul. 2022 – Jun. 2025

Invited Talks

2024, “Martes Talk” Colloquium, School of Astronomy and Space Science, Nanjing University, *“Turbulence Properties of Galactic Molecular Clouds Depicted by Column Density and Velocity Statistics”*, Nanjing, China.

Honors and Awards

2023 Excellent Presentation Award for Young Scholars, Star Formation and Molecular Clouds Conference
2017-2019 Merit student, University of Chinese Academy of Sciences
2014 Outstanding Student Leader, University of Chinese Academy of Sciences

Skills

Programming: Python, IDL, GILDAS, familiar with C
Data Analysis: spectral line reduction and analysis
Signal Processing: Fourier analysis (structure function, power spectrum, wavelet method)
Bayesian Inference: MCMC modeling and posterior analysis (e.g., emcee)
Simulation Data Handling: conversion between HDF5 and FITS; analysis of simulation cubes

Outreach

2014-2016 President of Student Union, PMO, CAS

2014-2015 Student Leader of Popular Science Association, PMO, CAS

Refereed Journal Papers

First/Corresponding Author Papers

- [1] Suziye He, **Ma, Yuehui**, Hongchi Wang, Renjie Shen, Miaomiao Zhang, Chong Li, Zhenyi Yue, Xiangyu Ou, and Xuepeng Chen. “Hierarchical Structure and Self-gravity in the Rosette Molecular Cloud”. In: ApJ 996.1, 62 (Jan. 2026), p. 62. DOI: [10.3847/1538-4357/ae1cc8](https://doi.org/10.3847/1538-4357/ae1cc8). arXiv: [2511.02334](https://arxiv.org/abs/2511.02334) [astro-ph.GA].
- [2] **Ma, Yuehui**, Miaomiao Zhang, Hongchi Wang, Min Fang, Zhenyi Yue, Xuepeng Chen, Ji Yang, Fujun Du, Yang Su, Suziye He, Haoran Feng, Yan Sun, Chong Li, Qing-Zeng Yan, Zhiwei Chen, Shaobo Zhang, and Xin Zhou. “Examining Turbulence in Galactic Molecular Clouds. I. A Statistical Analysis of Velocity Structures”. In: ApJ 979.1, 65 (Jan. 2025), p. 65. DOI: [10.3847/1538-4357/ad9b0e](https://doi.org/10.3847/1538-4357/ad9b0e). arXiv: [2501.11859](https://arxiv.org/abs/2501.11859) [astro-ph.GA].
- [3] Renjie Shen, **Ma, Yuehui**, Hongchi Wang, Suziye He, and Miaomiao Zhang. “Hierarchical Structure and Self-gravity in the Maddalena Giant Molecular Cloud”. In: ApJ 971.1, 14 (Aug. 2024), p. 14. DOI: [10.3847/1538-4357/ad5347](https://doi.org/10.3847/1538-4357/ad5347). arXiv: [2405.19766](https://arxiv.org/abs/2405.19766) [astro-ph.GA].
- [4] **Ma, Yuehui**, Hongchi Wang, Miaomiao Zhang, Chen Wang, Shaobo Zhang, Yao Liu, Chong Li, Yuqing Zheng, Lixia Yuan, and Ji Yang. “Gas Column Density Distribution of Molecular Clouds in the Third Quadrant of the Milky Way”. In: ApJS 262.1, 16 (Sept. 2022), p. 16. DOI: [10.3847/1538-4365/ac7797](https://doi.org/10.3847/1538-4365/ac7797). arXiv: [2206.03963](https://arxiv.org/abs/2206.03963) [astro-ph.GA].
- [5] **Ma, Yuehui**, Hongchi Wang, Chong Li, Lianghao Lin, Yan Sun, and Ji Yang. “Molecular Clouds in the Second Quadrant of the Milky Way Midplane from $l = 104^{\circ}75$ to $l = 119^{\circ}75$ and $b = -5^{\circ}25$ to $b = 5^{\circ}25$ ”. In: ApJS 254.1, 3 (May 2021), p. 3. DOI: [10.3847/1538-4365/abe85c](https://doi.org/10.3847/1538-4365/abe85c). arXiv: [2102.12199](https://arxiv.org/abs/2102.12199) [astro-ph.GA].
- [6] **Ma, Yuehui**, Hong-Chi Wang, Chong Li, and Ji Yang. “The column density structure of Orion A depicted by N-PDF”. In: Research in Astronomy and Astrophysics 20.4, 060 (Apr. 2020), p. 060. DOI: [10.1088/1674-4527/20/4/60](https://doi.org/10.1088/1674-4527/20/4/60). arXiv: [1912.06984](https://arxiv.org/abs/1912.06984) [astro-ph.GA].
- [7] **Ma, Yuehui**, Hongchi Wang, Miaomiao Zhang, Chong Li, and Ji Yang. “A Large-field $J = 1 - 0$ Survey of CO and Its Isotopologues toward the Cassiopeia A Supernova Remnant”. In: ApJ 878.1, 44 (June 2019), p. 44. DOI: [10.3847/1538-4357/ab1ea7](https://doi.org/10.3847/1538-4357/ab1ea7). arXiv: [1905.10193](https://arxiv.org/abs/1905.10193) [astro-ph.GA].

Co-authored Papers

- [8] Ji Yang, Qing-Zeng Yan, Yang Su, Shaobo Zhang, Xin Zhou, Yan Sun, Yiping Ao, Xuepeng Chen, Zhiwei Chen, Fujun Du, Min Fang, Yan Gong, Zhibo Jiang, Shengyu Jin, Binggang Ju, Chong Li, Yingjie Li, Yi Liu, Dengrong Lu, Chunsheng Luo, **Ma, Yuehui**, Ruiqing Mao, Jixian Sun, Chen Wang, Hongchi Wang, Min Wang, Min Wang (Qinghai), Xindong Wang, Wenting Xu, Ye Xu, Kun Yan, Ping Yan, Lixia Yuan, Miaomiao Zhang, and Yongxing Zhang. “The Milky Way Imaging Scroll Painting Survey: Data Release 1”. In: *ApJS* 282.2, 65 (Feb. 2026), p. 65. DOI: [10.3847/1538-4365/ae29e7](https://doi.org/10.3847/1538-4365/ae29e7). arXiv: [2512.08260](https://arxiv.org/abs/2512.08260) [[astro-ph.GA](#)].
- [9] Yang Su, Shiyu Zhang, Yan Sun, Ji Yang, Fujun Du, Min Fang, Qing-Zeng Yan, Shaobo Zhang, Zhiwei Chen, Xuepeng Chen, Xin Zhou, Lixia Yuan, and **Ma, Yuehui**. “Gas Transfer between the Inner 3 kpc Disk and the Galactic Central Molecular Zone”. In: *ApJ* 984.2, 109 (May 2025), p. 109. DOI: [10.3847/1538-4357/adc38e](https://doi.org/10.3847/1538-4357/adc38e). arXiv: [2503.12313](https://arxiv.org/abs/2503.12313) [[astro-ph.GA](#)].
- [10] Haoran Feng, Zhiwei Chen, Zhibo Jiang, **Ma, Yuehui**, Yang Yang, Shuling Yu, Dongqing Ge, Wei Zhou, Fujun Du, Chen Wang, Shiyu Zhang, Yang Su, and Ji Yang. “Revisiting the Velocity Dispersion–Size Relation in Molecular Cloud Structures”. In: *Research in Astronomy and Astrophysics* 24.11, 115018 (Nov. 2024), p. 115018. DOI: [10.1088/1674-4527/ad89a9](https://doi.org/10.1088/1674-4527/ad89a9). arXiv: [2410.15034](https://arxiv.org/abs/2410.15034) [[astro-ph.GA](#)].
- [11] Yan Sun, Ji Yang, Qing-Zeng Yan, Shaobo Zhang, Yang Su, Xuepeng Chen, Xin Zhou, **Ma, Yuehui**, and Lixia Yuan. “Molecular Clouds in the Outer Milky Way Disk: Sample, Integrated Properties, and Radial Trends with Galactocentric Radius”. In: *ApJS* 275.2, 35 (Dec. 2024), p. 35. DOI: [10.3847/1538-4365/ad8237](https://doi.org/10.3847/1538-4365/ad8237).
- [12] Shiyu Zhang, Yang Su, Xuepeng Chen, Min Fang, Qing-Zeng Yan, Shaobo Zhang, Yan Sun, Xiaolong Wang, Haoran Feng, **Ma, Yuehui**, Miaomiao Zhang, Zi Zhuang, Xin Zhou, Zhiwei Chen, and Ji Yang. “The Multilayer Nature of Molecular Gas toward the Cygnus Region”. In: *AJ* 167.5, 220 (May 2024), p. 220. DOI: [10.3847/1538-3881/ad2fcb](https://doi.org/10.3847/1538-3881/ad2fcb). arXiv: [2403.00061](https://arxiv.org/abs/2403.00061) [[astro-ph.GA](#)].
- [13] Chong Li, Keping Qiu, Di Li, Hongchi Wang, Yue Cao, Junhao Liu, **Ma, Yuehui**, and Chenglin Yang. “H I Self-absorption toward the Cygnus X North: From Atomic Filament to Molecular Filament”. In: *ApJL* 948.2, L17 (May 2023), p. L17. DOI: [10.3847/2041-8213/accf99](https://doi.org/10.3847/2041-8213/accf99). arXiv: [2305.10795](https://arxiv.org/abs/2305.10795) [[astro-ph.GA](#)].
- [14] Chunxue Li, Hongchi Wang, **Ma, Yuehui**, Lianghao Lin, Yang Su, Chong Li, Yan Sun, Xin Zhou, and Ji Yang. “Molecular Clouds in the Galactic Plane from $l = [59.^{\circ}75, 74.^{\circ}75]$ and $b = [-5.^{\circ}25, +5.^{\circ}25]$ ”. In: *ApJS* 267.2, 30 (Aug. 2023), p. 30. DOI: [10.3847/1538-4365/acd9a7](https://doi.org/10.3847/1538-4365/acd9a7). arXiv: [2305.18043](https://arxiv.org/abs/2305.18043) [[astro-ph.GA](#)].
- [15] Chen Wang, Haoran Feng, Ji Yang, Xuepeng Chen, Yang Su, Qing-Zeng Yan, Fujun Du, **Ma, Yuehui**, and Jiajun Cai. “The Molecular Clouds in a Section of the Third Galactic Quadrant: Observational Properties and Chemical Abundance Ratio between CO and its Isotopologues”. In: *AJ* 166.3, 121 (Sept. 2023), p. 121. DOI: [10.3847/1538-3881/acebdd](https://doi.org/10.3847/1538-3881/acebdd). arXiv: [2308.10726](https://arxiv.org/abs/2308.10726) [[astro-ph.GA](#)].

- [16] Chen Wang, Haoran Feng, Ji Yang, Xuepeng Chen, Yang Su, Qinq-Zeng Yan, Fujun Du, **Ma, Yuehui**, and Jiajun Cai. “The Molecular Clouds in a Section of the Third Galactic Quadrant: Cloud Catalog”. In: *AJ* 165.3, 106 (Mar. 2023), p. 106. DOI: [10.3847/1538-3881/acafee](https://doi.org/10.3847/1538-3881/acafee).
- [17] Lixia Yuan, Ji Yang, Fujun Du, Xunchuan Liu, Yang Su, Qing-Zeng Yan, Xuepeng Chen, Yan Sun, Shaobo Zhang, Xin Zhou, and **Ma, Yuehui**. “On the Spatial Distribution of ^{13}CO Structures within ^{12}CO Molecular Clouds”. In: *ApJ* 944.1, 91 (Feb. 2023), p. 91. DOI: [10.3847/1538-4357/acac26](https://doi.org/10.3847/1538-4357/acac26). arXiv: [2212.02066](https://arxiv.org/abs/2212.02066) [[astro-ph.GA](#)].
- [18] Xin Zhou, Yang Su, Ji Yang, Xuepeng Chen, Yan Sun, Zhibo Jiang, Min Wang, Hongchi Wang, Shaobo Zhang, Ye Xu, Qingzeng Yan, Lixia Yuan, Zhiwei Chen, Yiping Ao, and **Ma, Yuehui**. “A Systematic Study of Associations between Supernova Remnants and Molecular Clouds”. In: *ApJS* 268.2, 61 (Oct. 2023), p. 61. DOI: [10.3847/1538-4365/acee7f](https://doi.org/10.3847/1538-4365/acee7f). arXiv: [2308.03484](https://arxiv.org/abs/2308.03484) [[astro-ph.GA](#)].
- [19] Chun-Xue Li, Hong-Chi Wang, **Ma, Yue-hui**, Miao-Miao Zhang, Chong Li, and Yu-Qing Zheng. “Molecular Clouds Associated with H II Regions and Candidates within $l = 106.^{\circ}65$ to $109.^{\circ}50$ and $b = -1.^{\circ}85$ to $0.^{\circ}95$ ”. In: *Research in Astronomy and Astrophysics* 22.4, 045008 (Apr. 2022), p. 045008. DOI: [10.1088/1674-4527/ac52a0](https://doi.org/10.1088/1674-4527/ac52a0). arXiv: [2202.04264](https://arxiv.org/abs/2202.04264) [[astro-ph.GA](#)].
- [20] Jia Sun, Robert A. Gutermuth, Hongchi Wang, Miaomiao Zhang, Shuinai Zhang, **Ma, Yuehui**, Xinyu Du, and Min Long. “Deep near-infrared survey towards the W40 and Serpens South region in the Aquila Rift: A comprehensive catalogue of young stellar objects”. In: *MNRAS* 516.4 (Nov. 2022), pp. 5244–5257. DOI: [10.1093/mnras/stac2191](https://doi.org/10.1093/mnras/stac2191). arXiv: [2207.09041](https://arxiv.org/abs/2207.09041) [[astro-ph.SR](#)].
- [21] Lixia Yuan, Ji Yang, Fujun Du, Yang Su, Xunchuan Liu, Shaobo Zhang, Yan Sun, Xin Zhou, Qing-Zeng Yan, and **Ma, Yuehui**. “Molecular Gas Structures Traced by ^{13}CO Emission in the 18,190 ^{12}CO Molecular Clouds from the MWISP Survey”. In: *ApJS* 261.2, 37 (Aug. 2022), p. 37. DOI: [10.3847/1538-4365/ac739f](https://doi.org/10.3847/1538-4365/ac739f). arXiv: [2205.14858](https://arxiv.org/abs/2205.14858) [[astro-ph.GA](#)].
- [22] Lixia Yuan, Ji Yang, Fujun Du, Xunchuan Liu, Shaobo Zhang, Zehao Lin, Jingfei Sun, Qing-Zeng Yan, **Ma, Yuehui**, Yang Su, Yan Sun, and Xin Zhou. “A Morphological Classification of 18,190 Molecular Clouds Identified in ^{12}CO Data from the MWISP Survey”. In: *ApJS* 257.2, 51 (Dec. 2021), p. 51. DOI: [10.3847/1538-4365/ac242a](https://doi.org/10.3847/1538-4365/ac242a). arXiv: [2108.12110](https://arxiv.org/abs/2108.12110) [[astro-ph.GA](#)].
- [23] Yu-Qing Zheng, Hong-Chi Wang, **Ma, Yue-hui**, and Chong Li. “Symmetry properties and widths of the filamentary structures in the Orion A giant molecular cloud”. In: *Research in Astronomy and Astrophysics* 21.8, 188 (Oct. 2021), p. 188. DOI: [10.1088/1674-4527/21/8/188](https://doi.org/10.1088/1674-4527/21/8/188). arXiv: [2103.10637](https://arxiv.org/abs/2103.10637) [[astro-ph.GA](#)].
- [24] Chakali Eswaraiah, Di Li, Manash R. Samal, Jia-Wei Wang, **Ma, Yuehui**, Shih-Ping Lai, Annie Zavagno, Tao-Chung Ching, Tie Liu, Kate Pattle, Derek Ward-Thompson, Anil K. Pandey, and Devendra K. Ojha. “Unveiling the Importance of Magnetic Fields in the Evolution of Dense Clumps Formed at the Waist of Bipolar H II Regions: A Case Study of Sh 2-201 with JCMT SCUBA-2/POL-2”. In: *ApJ* 897.1, 90 (July 2020), p. 90. DOI: [10.3847/1538-4357/ab83f2](https://doi.org/10.3847/1538-4357/ab83f2). arXiv: [1912.10188](https://arxiv.org/abs/1912.10188) [[astro-ph.GA](#)].

- [25] Chong Li, Hong-Chi Wang, Yuan-Wei Wu, **Ma, Yue-hui**, and Liang-Hao Lin. “Assessing the performance of molecular gas clump identification algorithms”. In: *Research in Astronomy and Astrophysics* 20.3, 031 (Mar. 2020), p. 031. DOI: [10.1088/1674-4527/20/3/31](https://doi.org/10.1088/1674-4527/20/3/31). arXiv: [1910.07692](https://arxiv.org/abs/1910.07692) [[astro-ph.GA](#)].
- [26] Chong Li, Hongchi Wang, Miaomiao Zhang, **Ma, Yuehui**, and Lianghao Lin. “The Molecular Clouds Associated with the H II Regions/Candidates between $L = 207^\circ$ and $L = 211^\circ$ ”. In: *ApJS* 249.2, 27 (Aug. 2020), p. 27. DOI: [10.3847/1538-4365/ab9cc0](https://doi.org/10.3847/1538-4365/ab9cc0).
- [27] Chakali Eswaraiah, Shih-Ping Lai, **Ma, Yuehui**, Anil K. Pandey, Jessy Jose, Zhiwei Chen, Manash R. Samal, Jia-Wei Wang, Saurabh Sharma, and D. K. Ojha. “Polarimetric and Photometric Investigation of the Dark Globule LDN 1225: Distance, Extinction Law, and Magnetic Fields”. In: *ApJ* 875.1, 64 (Apr. 2019), p. 64. DOI: [10.3847/1538-4357/ab0a0c](https://doi.org/10.3847/1538-4357/ab0a0c). arXiv: [1812.08920](https://arxiv.org/abs/1812.08920) [[astro-ph.GA](#)].
- [28] Yingjie Li, Ye Xu, Yan Sun, Qing-Zeng Yan, **Ma, Yuehui**, and Ji Yang. “CO Outflow Candidates toward the W3/4/5 Complex I: The Sample and Its Spatial Distribution”. In: *ApJS* 242.2, 19 (June 2019), p. 19. DOI: [10.3847/1538-4365/ab1e55](https://doi.org/10.3847/1538-4365/ab1e55). arXiv: [1905.02018](https://arxiv.org/abs/1905.02018) [[astro-ph.GA](#)].
- [29] Chen Wang, Ji Yang, Yang Su, Fujun Du, **Ma, Yuehui**, and Shaobo Zhang. “Molecular Gas toward the Gemini OB1 Molecular Cloud Complex. III. Chemical Abundance”. In: *ApJS* 243.2, 25 (Aug. 2019), p. 25. DOI: [10.3847/1538-4365/ab2d2e](https://doi.org/10.3847/1538-4365/ab2d2e).
- [30] Chong Li, Hongchi Wang, Miaomiao Zhang, **Ma, Yuehui**, Min Fang, and Ji Yang. “A Large-scale Survey of CO and Its Isotopologues toward the Rosette Molecular Cloud”. In: *ApJS* 238.1, 10 (Sept. 2018), p. 10. DOI: [10.3847/1538-4365/aad963](https://doi.org/10.3847/1538-4365/aad963). arXiv: [1808.09189](https://arxiv.org/abs/1808.09189) [[astro-ph.GA](#)].